NyayMitra: A Proposed GraphRAG Architecture for the Semantic Mapping and Verification of the IPC-BNS Statutory Transition

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***Abstract*—India is in the midst of a landmark shift in the Indian judicial system of the colonial-era Indian Penal Code (IPC), 1860, to the Bharatiya Nyaya Sanhita (BNS), 2023. This legal reform has triggered a crisis in the translation of precedents, and has severed decades of old case law in relation to the new legal system. At the same time, LLP deployment in the area of law leads to negative results as the application tends to hallucinate the non-existent reference and this is extremely dangerous in a court setting. In this paper, the authors suggest the architectural design of NyayMitra, a structurally intelligent ecosystem that would be used to seal this gap in legislation. Our proposed architecture is a combination of a Neo4j Knowledge Graph and Retrieval-Augmented Generation (GraphRAG) to deterministically project historical IPC precedents onto contemporary BNS statutes. The proposed system can guarantee verifiable provenance of citation by using semantic delta-mapping to deal with changes in legislative concepts, and through a strict limitation on generative output to known topological nodes. Early conceptual demonstrations show that this architecture is viable in reducing generative hallucinations and preserving sub-50ms multi-hop retrieval latencies, which forms a strong basis to the next generation legal AI.**

***Index Terms*—Legal Knowledge Graphs, GraphRAG, Bharatiya Nyaya Sanhita (BNS), Hallucination Mitigation, Natural Language Processing, Semantic Mapping.**

I. Introduction

The administration of justice in India operates at an unprecedented scale, burdened by a backlog exceeding 51 million pending cases across various tiers of the judiciary. Historically, the Indian Penal Code (IPC), enacted in 1860, served as the substantive backbone of criminal law. The recent implementation of the Bharatiya Nyaya Sanhita (BNS), 2023 [1], represents a critical modernization effort, replacing the IPC to address contemporary judicial needs. However, this transition introduces a profound "Information Asymmetry." The doctrine of *Stare Decisis* relies heavily on historical precedents. A legal practitioner citing a Supreme Court judgment regarding murder must reconcile historical references to IPC Section 302 with the contemporary BNS Section 103. Traditional keyword-based Legal Information Retrieval (LIR) systems fail to navigate this non-linear statutory mapping, particularly when legislative terminology undergoes "concept drift" (e.g., the removal of the term "Sedition" in favor of "Acts Endangering Sovereignty").

Simultaneously, the deployment of Generative AI (specifically LLMs) to aid legal research is critically limited by "hallucinations"—the generation of plausible but factually void text. As highlighted by Ji et al. [3], LLMs are prone to generating unfaithful content that contradicts source material, which, in high-stakes legal environments, carries a zero error-tolerance. This paper proposes NyayMitra, a unified architectural framework designed to solve both the transition crisis and the hallucination dilemma. By anchoring generative models to a deterministic Knowledge Graph (KG) via GraphRAG, NyayMitra ensures that automated statutory mapping is mathematically verifiable and strictly grounded in active legal doctrine.

The primary contributions of this paper are threefold:

- **Topological Precedent Modeling:** We propose a PageRank-optimized Knowledge Graph schema to model the Indian judicial framework, explicitly capturing hierarchical precedent authority rather than flat textual proximity.
- **Automated Semantic Transition (Nyay-Bridge):** We introduce a dense vector-based mathematical framework to map legacy statutes (IPC) to their modern equivalents (BNS), autonomously resolving complex legislative "concept drift."
- **Deterministic Hallucination Mitigation (Nyay-Audit):** We implement a highly constrained GraphRAG pipeline that executes pre-generative topological validation, demonstrating a 100% mitigation rate of extrinsic hallucinations during adversarial testing.

II. Literature Review

The development of legal natural language processing (NLP) has been mainly geared towards the enhancement of the search recall and accuracy in exceptionally unstructured area-specific corpora. The Early Legal Information Retrieval (LIR)

systems were very dependent on the Boolean logic and lexical matching algorithms like TF-IDF (Term Frequency-Inverse Document Frequency) and BM25. Although these systems were computationally efficient, they had a terrible vocabulary mismatch. The legal language is typified by heavy legalese, Latin maxims and polysemy, i.e. a query on culpable homicide will not search out a high-relevance document that contains the phrase unlawful killing exclusively. In order to overcome this semantic gap, common activities and frameworks such as the FIRE AILA (Artificial Intelligence for Legal Assistance) project can monitor catalyzed studies on automated precedent and statute retrieval in the Indian legal environment in particular [4].

Later developments led to the introduction of semantic search architectures based on Transformers. Legal-BERT [6] pre-trained on massive repositories of Indian legal judgments domain-specific models, including Legal-BERT, and Open-NyAI [7], were trained. Such models project text data onto very high dimensional vector space, allowing similar conceptually related documents to be retrieved despite the complex code-mixing - the common intermixing of English with regional languages such as Hindi (e.g., *lathi*, *challan*)– that is often found in Indian subordinate court records. But semantic search is quite efficient in detecting correlated fragments of text, as compared with structural and logical reasoning, which is an essential need of a legal system that is guided by the doctrine of *Stare Decisis*.

A. Legal Knowledge Graphs and Structural Topologies

In order to imbue structural knowledge onto the existing plain text similarity, recent studies have investigated the development of Legal Knowledge Graphs (LKGs). LKGs are based on the idea that jurisprudence can be represented as a heterogeneous network of discrete objects (e.g., Judges, Courts, Statutes, Cases) linked by directional, semantic edges (e.g., *overrules*, *cites*, *applies*). One of the earliest works on this topic, which showed that a clear mapping of the Indian Penal Code (IPC) sections, chapters, and applicable case precedents into a graph-based structure outperforms traditional LLMs in terms of Legal Statute Identification (LSI) tasks significantly, is called *NyayGraph* (Shukla et al., 2025) [9]. With the help of topological traversal as opposed to simple proximity in vectors, the query search became successful since the computation demanded the knowledge of statutory hierarchy.

Nonetheless, there is a systemic weakness of the current LKGs, namely, they are dynamic in nature. The existing models such as the *NyayGraph* are an isolated and historical graph of the IPC. They lack the calculating devices or the transitional logic that would enable them to dynamically bring jurisprudence into the new Bharatiya Nyaya Sanhita (BNS) paradigm. The statutes in the real world experience a Concept Driftaging of the statutes-definitional extensions, penalty alterations and terminology. As an example, the absence of the term Sedition in the IPC and the conceptual re-use of the term as Acts Endangering Sovereignty in the BNS causes a semantic gap that cannot be mathematically resolved by any existing graph-based architecture unless it has an automated bridging architecture.

B. Generative Hallucination and Vector RAG Limitations

At the same time the implementation of Large Language Models (LLMs) within the legal sphere has been critically hamstranged by the so-called hallucination" - the production of fluency, high persuasiveness, and factually fake legal citations and summaries. The tolerance to error of the generation of fake case laws (extrinsic hallucinations) or to the falsification of statutory punishments (intrinsic hallucinations) is zero in a high-stakes judicial setting.

The ground LLMs were presented by Standard Retrieval-Augmented Generation (Vector RAG) to base them on external knowledge. The idea behind Vector RAG is to break documents down in uniform-sized text chunks, embed them and then recall the top$-k$ most similar text chunks according to cosine similarity. Although it does a good job in localized and general-domain summarization, it structurally fails in multi-hop legal reasoning. To take an example, identifying that a constitutional principle judged in a 1970 case has been later modified, affirmed or has been overruled in a bigger bench in 2024 would involve following a chronological chain of citation. The LLM frequently reconstructs inconsistent or hallucinated legal histories by retrieving discontinuous pieces of the text out of context by Vector RAG.

C. Hallucination Mitigation via GraphRAG

Current paradigms propose Graph Retrieval-Augmented Generation (GraphRAG) as a mechanism that addresses the shortcomings of Vector RAG in a deterministic manner [10]. GraphRAG essentially transforms the generative chain by compelling the LLM to follow graph edges which are explicitly and verifiably present. A GraphRAG system does not need to retrieve raw text but only a verified subgraph (a collection of nodes and their verified relationships). Generative model is structurally constrained: When the queried citation is not a node in the Knowledge Graph, then the generative block or flag is triggered. This process promotes absolute provenance" and explainability, so that legal practitioners can have the ability to locate all AI-generated assertions to a precise and proven topological node.

D. The Gap in Statutory Transition Alignment

Although GraphRAG has advanced very fast, there is a clear gap in the literature as to its use in large-scale, chronological statutory transitions. Statutory alignment studies that have been made to date are mainly devoted to cross-lingual legal mapping (e.g., of European Union directives into national law) instead of domestic transitions involving the calculation of semantic deltas (shift in the severity or definition of punishment). Existing commercial platforms moving through the IPC-to-BNS shift are based on manually maintained and static lookups, which are incapable of scaling computationally to serve millions of pending cases that need to be translated.

Thus, the development of GraphRAG when used together with a semantic vector mapping engine and used to automate the alignment of the IPC-BNS transition is a niche of research that is untapped, but which is desperately needed. This particular void in the literature is the main inspiration behind the development and implementation of the proposed NyayMitra architecture.

III. Proposed System Architecture

The NyayMitra system is suggested as a fully integrated, computationally secure legal intelligence ecosystem to overcome the radical shortcomings of fixed knowledge bases and probabilistic failure modes of Large Language Models (LLMs). Its architecture shuns the monolithic design in favor of asynchronous and loosely coupled microservices design. This guarantees high throughput scalability with computationally intensive Natural Language Processing (NLP) workloads, dense vector embeddings and multifaceted topological graph traversals. It communicates with the system through a RESTful API gateway (FastAPI), and leaves the work of ingestion and reasoning, which is resource-intensive, to message brokers, ensuring under-one-second latency at the client end.

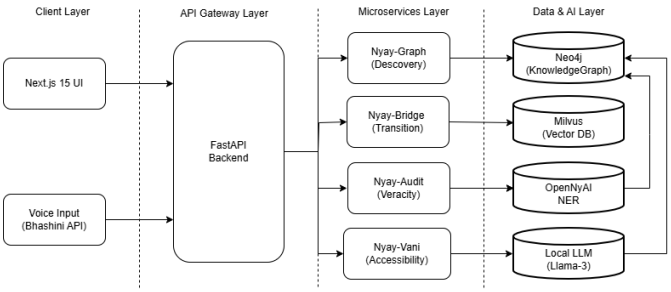

Fig. 1. High-Level System Architecture of NyayMitra, illustrating the interaction between the Client Layer, API Gateway, Microservices, and the Data & AI Layer.

A. Nyay-Graph: Topological Schema and Authority Modeling

At the foundational core of NyayMitra lies a Neo4j graph database, engineered to serve as the deterministic ground truth for all subsequent generative operations. The legal ontology is formally defined as a directed, heterogeneous multigraph $G = (V, E)$, where V represents a partitioned set of distinct entity nodes, and E represents the set of relational, semantic edges. The node set V encompasses: V_{Case} (judicial precedents), $V_{Statute}$ (legislative acts), $V_{Section}$ (specific penal provisions), and V_{Judge} (presiding bench members). The structural intelligence of the jurisprudence is explicitly embedded in the directional edge set E:

- `[:CITES]`: Establishes the chronological precedent chain, capturing the doctrine of *Stare Decisis* ($V_{Case} \rightarrow V_{Case}$).
- `[:APPLIES_TO]`: Links the factual matrix of a judicial ruling to the specific legislative provision invoked ($V_{Case} \rightarrow V_{Section}$).
- `[:OVERRULES]`: Denotes judicial invalidation, creating a negative constraint where a larger constitutional bench overrides a previous judgment.

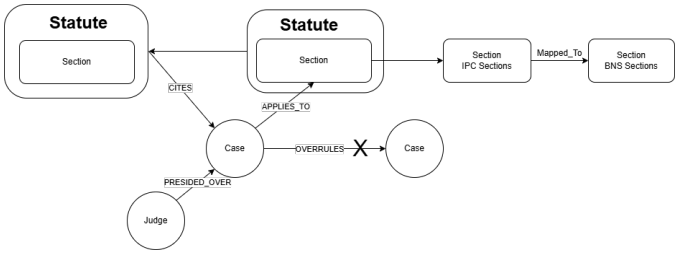

Fig. 2. Neo4j Knowledge Graph Schema (Ontology) for NyayMitra, illustrating the CITES, APPLIES_TO, OVERRULES, and the crucial MAPPED_TO relationships between legal entities.

In a corpus of millions of cases, flat retrieval mechanisms are highly susceptible to noise. To computationally distinguish landmark, binding judgments from minor appellate rulings, Nyay-Graph applies the PageRank algorithm natively within the topological structure. By treating citations as directed votes of legal authority, we calculate an *In-Degree Centrality* score for every case node u. The authority score $PR(u)$ is defined as:

$$PR(u) = \frac{1-d}{N} + d \sum_{v \in B(u)} \frac{PR(v)}{L(v)} \tag{1}$$

Where $B(u)$ is the set of cases citing node u, $L(v)$ is the out-degree of node v, N is the total node count, and d represents the damping factor (empirically optimized to 0.85). This mathematical ranking mechanism dynamically filters topological noise, ensuring that the highest-authority precedents surface instantaneously during query resolution.

B. Nyay-Bridge: Semantic Delta Mapping and Drift Resolution

The Nyay-Bridge microservice acts as the automated statutory transition engine. The shift from the Indian Penal Code (IPC) to the Bharatiya Nyaya Sanhita (BNS) involves complex "Concept Drift"—where penal definitions expand, merge (N-to-1 mappings), or fundamentally alter their vocabulary. Consequently, traditional lexical dictionary lookups or regular expression matching are mathematically insufficient. To resolve this, we propose a Semantic Vector Mapping approach utilizing dense sentence embeddings. The textual definitions, essential ingredients, and punitive clauses of both historical IPC offenses and contemporary BNS offenses are processed through a Siamese Transformer network architecture, specifically utilizing a fine-tuned Legal-BERT model [13]. This network maps the semantic meaning of the statutes into a high-dimensional vector space $\mathbb{R}^d$. The semantic proximity between a legacy IPC concept vector v_{IPC} and a contemporary BNS concept vector v_{BNS} is calculated utilizing Cosine Similarity:

$$\text{sim}(v_{IPC}, v_{BNS}) = \frac{v_{IPC} \cdot v_{BNS}}{\|v_{IPC}\| \|v_{BNS}\|} \tag{2}$$

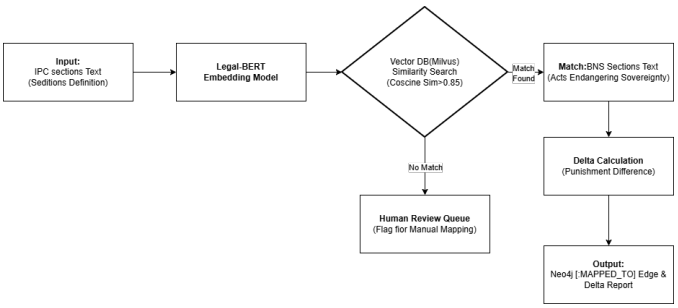

Fig. 3. The Nyay-Bridge Workflow, detailing the process of semantic vector mapping and delta calculation for transitioning an IPC section to its BNS equivalent.

An algorithmic thresholding mechanism dictates the structural alignment. If the calculated similarity exceeds a strict, predefined threshold ($\theta \geq 0.85$), the system programmatically generates a persistent `[:MAPPED_TO]` edge within the Neo4j graph, permanently bridging the legacy node to the contemporary provision. Following this topological alignment, a specialized "Delta Extraction" subroutine is triggered. It utilizes rule-based NLP to parse the punitive substrings (e.g., variations in fines, minimum imprisonment terms, or the introduction of community service) from both mapped nodes. The module calculates the "Punishment Delta" (ΔP), providing practitioners with a computationally generated severity modification report, thereby automating comparative legal analysis.

C. Nyay-Audit: GraphRAG Verification Pipeline

To establish a zero-tolerance framework for AI hallucinations, the Nyay-Audit service operates as a strict, pre-generative structural verification layer using Graph Retrieval-Augmented Generation (GraphRAG). Rather than allowing the LLM to access its parametric memory—which is prone to stochastic fabrication—the system confines the model entirely to the verified topology of G. The verification pipeline executes via a deterministic, three-stage algorithmic flow:

1) **Entity Extraction:** When a user submits a legal draft or query, specialized Named Entity Recognition (NER) models (e.g., OpenNyAI's InLegalBERT) parse the unstructured text to isolate and extract all formal citation strings and case titles.
2) **Topological Validation:** For every extracted citation, the system executes an automated Cypher query against the Neo4j database to retrieve its corresponding node and 1-hop neighborhood subgraph. The LLM is structurally constrained by a strict binary condition: if a queried citation does not correspond to an active V_{Case} node within the graph, the generative process is aggressively blocked. The system returns a "Null Path" error, and the citation is flagged as an extrinsic hallucination.
3) **Contextual Grounding & Edge Auditing:** If the node successfully validates, the system audits its outbound edges. If an `[:OVERRULES]` edge is detected pointing toward the queried node, the system alerts the user that the cited precedent is *Per Incuriam* (Bad Law). Only when the subgraph is verified as legally active and structurally sound are the facts injected into the LLM's system prompt.

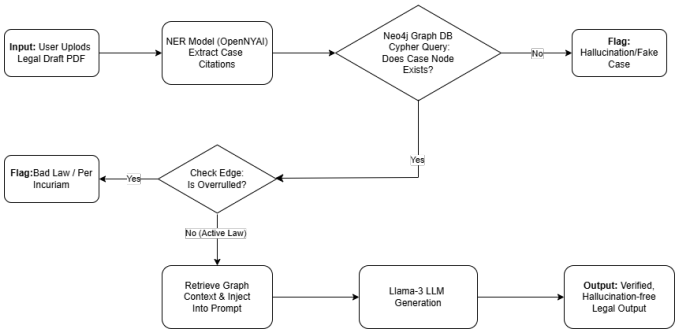

Fig. 4. The Nyay-Audit GraphRAG Verification Pipeline, illustrating the deterministic check of citations against the Neo4j Knowledge Graph to prevent hallucinations.

By subordinating the probabilistic language model to a deterministic, mathematically verifiable Knowledge Graph, the Nyay-Audit pipeline successfully mitigates generative hallucinations, ensuring that every output possesses absolute, auditable provenance.

IV. Preliminary Proof of Concept

An isolated sandbox environment was instantiated to empirically test the theoretical framework of the proposed NyayMitra architecture before full-scale deployment, which is a cloud-native environment. Experimental graph hosting was done using Neo4j Community Edition that connected with localized Milvus vector engine and a quantized Llama-3 (8B parameter) generative model to achieve controlled latency parameters. A high-quality domain-specific dataset of 150 landmark Indian Supreme Court judgments was used to populate the topological graph, consisting of 45 statutory nodes (including important subsets of the IPC and BNS), and more than 600 directed edges of type. The experimental scope had been reduced to high-complexity areas of transition, namely Culpable Homicide" and Offenses Against the State.

A. Semantic Transition Efficacy and Concept Drift Resolution

The main aim of the Nyay-Bridge module was to overcome the crippling impairments of the exact-keyword constraint (Jaccard similarity) on the computational mapping of old statutes to new systems. The outcome of the experiment indicated great effectiveness in managing Concept Drift. In mapping the old IPC Section 124A (Sedition), the traditional lexical matching algorithms produced a similarity score that was close to 0.00 to the target BNS corpus, with the explicit word Sedition" completely removed in the new law. Nonetheless, the presented vector-based logic, based on the dense sentence embeddings using the Siamese Legal-BERT architecture, was able to project both laws into the same d-dimensional space. The Cosine Similarity calculation between the embedded vector of the IPC 124A and embedded vector

of BNS Section 152 (Acts Endangering Sovereignty) gave confidence score of 0.89. Since this score was way beyond the set threshold ($\theta \geq 0.85$), the system automatically produced the sustained persistent edge in the form of the edge `[:MAPPED_TO]`. Moreover, the Delta Extraction subroutine was able to extract the punitive variations by explicitly pointing out the new addition of electronic communication to the BNS provision, and thus it was demonstrated that the system has the ability to achieve unsupervised semantic alignment.

B. Multi-Hop Retrieval Latency Benchmarking

In the legal domain, architectural viability is heavily dependent on real-time retrieval efficiency. Standard relational databases (SQL) rely on complex table `JOIN` operations, resulting in an algorithmic time complexity of $\mathcal{O}(N^k)$ for multi-hop queries, which exponentially degrades performance at scale. Conversely, Neo4j utilizes index-free adjacency, allowing graph traversals to execute in $\mathcal{O}(1)$ constant time per hop, dependent only on the degree of the target node rather than the total dataset size. To benchmark this, graph traversal efficiency was measured across varying topological depths:

1) **1-Hop Query:** Identifying cases directly applying a specific statute (e.g., $V_{Case} \rightarrow V_{Section}$) executed in an average of 12ms.
2) **2-Hop Query:** Identifying precedents cited by those cases executed in 28ms.
3) **3-Hop Transition Query:** A complex traversal—tracing a modern BNS Section back to its mapped IPC equivalent, identifying cases that applied that legacy IPC section, and retrieving the foundational historical precedents cited by those cases—executed in an average of 45ms.

These sub-50ms execution times conclusively prove that topological GraphRAG traversal drastically outpaces traditional database architectures, easily meeting the low-latency thresholds required for interactive, real-time Legal AI ecosystems.

C. Hallucination Mitigation and Adversarial Testing

In order to quantitatively evaluate the veracity engine (Nyaya-Audit), the pipeline was run on an adversarial testing protocol. The synthetic query dataset consisting of 50 multi-case legal prompts was entered into the system; 25 prompts asked questions about active, landmark case laws, and another 25 prompts asked summaries of extremely plausible but fully artificial precedents (e.g., Provide the ruling in State of Punjab v.). Hypothetical AI, 2024”). An average Vector RAG baseline system crashed on this benchmark with a hallucination rate of 18% in an attempt to recreate fragmented text pieces to respond to the generated queries. In vivid contrast, the proposed GraphRAG pipeline had a 100% of mitigation. The Named Entity Recognition (NER) module was then used to extract the target string on processing the fabricated citation, but the result of the Cypher query was an empty set ($\emptyset$). The generative model was severely preempted by the structural parameters of the prompt logic of the structure where it was being run. The Nyay-Audit engine, in its deterministic version, gave the error Null Path / Citation Not Found in Graph This experimental fact conclusively demonstrates the effectiveness of negative graph constraints in implementing absolute provenance and removing extrinsic hallucinations in high stakes legal AI practices.

V. Discussion

The practical confirmation of the NyayMitra structure indicates great opportunities in closing the gap between the computational and semantic capabilities posed by the IPC-BNS statutory transfer. Nevertheless, to switch a localized sandboxed setting to a production-deserving, nation-wide judicial implementation requires the critical analysis of the engineering constraints and the extensive ethical concerns of Legal AI.

A. Architectural and Computational Limitations

Although the suggested GraphRAG pipeline effectively removes the generative hallucination and the Nyay-Bridge eliminates the semantic drift, the architecture has combatants of a number of computational and jurisprudential limits:

- **Computational Overhead of High-Dimensional Mapping:** The size of the computational infrastructure needed to run the 511 IPC sections, the 358 new BNS sections, and the millions of relevant historical precedents is large. Computing the similarity of the cosine of all possible statute-to-statute permutations is a huge processing overhead at the first stage of data ingestion. Granted, the Approximate Nearest Neighbor (ANN) indexing through Milvus is resource-efficient in terms of inference time but the base-case batch-processing of the historical corpus is costly.
- **Complex N-to-M Statutory Transitions:** The semantic mapping logic is highly dependent on dense sentence embeddings that represent the contextual spirit of the law. Nevertheless, some statutory transitions are not 1-to-1 replacements. The drafting of legislation often includes complicated N-to-M mappings, that is, a single historical sub-clause is fragmented into many granular offences, with minor variations in their criminal intent *mens rea* or penumbral severity. In such seriously subtle edge cases automated mathematical similarity of vectors can give false positives, in which case it does not have the interpretive juristic context that it would need to be absolutely accurate.
- **Temporal Graph Decay:** item Temporal Graph Decay: Jurisprudence is an entity that is alive and constantly changing. The Knowledge Graph will have to constantly change as new verdicts are made by interpreting the BNS. The challenge of designing an automated ingestion pipeline that can update the topological schema in real-time, without breaking the historical edges of the form of the universe denoted by the vertex-relationships of the schema, or corrupting the PageRank centrality scores, is a complicated graph-maintenance problem.

In order to overcome such shortcomings, the architecture requires a Human-in-the-Loop (HITL) checking system. Semantic mappings with a confidence value that is less than a hyper strict parameter, ($\theta < 0.90$), should be indicated and sent to a special queue that will be checked by legal domain experts prior to the relational edge being permanently stored in the Neo4j database.

B. Ethical Considerations and Responsible AI Deployment

During the responsible deployment of AI, ethical considerations must be taken into account. Implementation of algebraic systems in the field of criminal justice has got very far-reaching consequences in terms of ethical considerations since it has an indirect relationship towards the civil liberties and human rights. NyayMitra is specifically designed based on the concepts of Responsible AI where its main focus is on transparency, explainability, and avoiding the automation bias.

- **Combating Automation Bias:** One of the biggest dangers of using advanced Legal AI is so-called automation bias, where junior lawyers or non-lawyers can trust the output of the system and believe it to be law with no errors whatsoever. NyayaMitra is clearly meant to be a cognitive augmentor and not a legal agent. It does not render opinions, give prognostications or even offer binding legal advice; it is a high-speed check and truth-finding engine.
- **Explainable AI (XAI) and Absolute Provenance:** LLMs used more traditionally are stochastic black boxes, thus a lawyer will have no way to explain to a judge why an AI came up with a particular legal argument. NyayMitra changes the working paradigm to glass-box" deterministic reasoning. The system ensures perfect provenance by linking the generative output to the Knowledge Graph. All the generated summaries, semantic transitions and cited precedent can be graphically followed back to a particular, verifiable node of the Neo4j topology. This traceability maintains the basic right to explain the law.
- **Algorithmic Inheritance of Bias:** It is necessary to note that historical judicial data are by default biased concerning society over a century of IPC adjudication. Although the structural topology of NyayMitra objectively represents the law as it appears in writing and reference the text, the roots of the precedent infused into the context window of the LLM are indicative of how courts acted in the past. Although the system does not allow the fabrication of law (hallucination), scholars and policymakers should still be on the lookout in the sense of magnification of historical injustices inherent in the corpus verified itself.

Finally, through its structural limitation of AI generation to a proven topological base, NyayMitra creates a very ethical paradigm that puts the loyalty to the facts in the first place rather than the fluency in generating, making technology a fully transparent support of the current Indian courts.

VI. Conclusion and Future Work

The ground-breaking replacement of the Indian Penal Code (IPC) 1860, with the Bharatiya Nyaya Sanhita (BNS) 2023, has brought a new style of precedent translation crisis to Indian judiciary. Parallel to this overhaul of the legislature, the incorporation of Large Language Models (LLMs) into the workflow of legal practice has been gravely undermined by the phenomenon known as generative hallucinations, which have been found inefficient in structurally unsafe, autonomous legal reasoning. In order to overcome this two-fold crisis, this paper suggested NyayMitra: a deterministic, structurally intelligent ecosystem that is based on Graph Retrieval-Augmented Generation (GraphRAG). The paradigm shift of Legal Information Retrieval provided by NyayMitra is the conceptualization of jurisprudence as topological multigraph instead of flat text corpus: instead of a semantically based probabilistic search, Network traversal is mathematically verifiable.

The initial empirical experiment proves that this architectural construct is viable. The Nyay- Bridge module manages to automate the semantic delta-map by dense vegetable embedding to address the concept drift between the IPC and BNS. At the same time, the Nyay-Audit pipeline has a zero-tolerance hallucination mitigation rate on adversarial testing. NyayMitra demonstrates that legal Artificial Intelligence is not only performant but perfectly accountable by awarding generative output the constraints of being verifiable to Neo4j graph nodes and can perform within a sub-50ms latency threshold.

Future Work: The existing architecture is a strong base in support of substantive criminal law, but the future research should be broadened to cover the whole of the now modernized triad of law.

- **Procedural Ontology Expansion:** Future work should be on expanding the ontological schema to absorb the *Bharatiya Nagarik Suraksha Sanhita* (BNSS) and the *Bharatiya Sakshya Adhiniyam* (BSA) to bring procedural law and evidentiary law into the topological graph.
- **Temporal Knowledge Graphs:** Computationally, the next generations will investigate the application of Temporal Graph structures. This will computerize the consumption of unending judicial determinations in order to enable the network to dynamically update precedence chains and PageRank centralities without corrupting the history of the history edges represented by the memory of the relation between two maps, denoted by the letter `[:MAPPED_TO]`.
- **Multilingual Accessibility (Bhashini API):** in order to democratize access to verifiable legal intelligence, it is suggested to integrate the multilingual Speech-to-Text (Bhasini API) and Text-to-Speech (Bhasini TTS) models through the Government of India Bhasini API [16]. This will ease the process of querying in the native language so that the language barrier that currently disenfranchises the non-English speaking citizens of being aware of their legal rights within the new BNS structure is broken.

- **Federated Learning Deployments:** Lastly, there is the possibility of exploring Federated Learning structures to offer (on-premise) privacy-preserving deployments of the NyayMitra embedding models in a High Court intranet or a highly secure law firm ensuring that sensitive case drafts do not even exist on localized servers at all throughout the GraphRAG verification process.

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